Laboratory 3: Counting Statistics and Error Analysis

Advance Reading

Knoll Chapter 3



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Information

- Name:
- Lab Station:
- Lab Partner:
- Date:

Theory Guiding Questions

- What is the binomial distribution and how is it derived? How is it relevant to counting statistics?
- What are the Gaussian and Poisson distributions and how are they related to each other and the binomial distribution?
- What is the chi-squared parameter and how does it describe a given dataset?

Pre-Lab Conceptual Questions

1. What is the relationship between the Gaussian and Poisson distribution? What are the criteria for using each one?
2. You took a single measurement with a detector and recorded 10,000 counts. What is the best estimate of the uncertainty and variance of this value assuming:
   (a) A Poisson distribution
   (b) A Gaussian distribution
3. Beginning with the error propagation formula, derive the uncertainty of final value w in the following equation where x, y, and z are measured values and μ_1 and μ_2 are constant. $w = z\,exp\left(-\left(\mu_1 x + \mu_2 y\right)\right)$

4. Derive “Currie Equation” with 99% confidence. For example, if somebody measured the no-source background and recorded 100 counts during a 2-min measurement, how should he/she set up the critical level and what is MDA with 99% confidence? We assume that the measurement time with an unknown source is also 2 mins.

2 The Single-Channel Analyzer